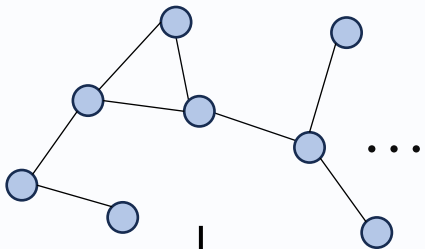
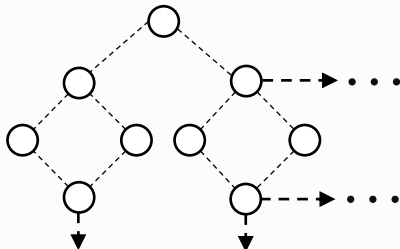


1 Graph Information Extraction with LLM

Raw Graph Data



Structural Template



Node-level Text Description



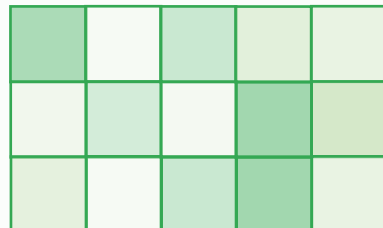
2 LLM-Based Semantic Embedding

LLM Encoder
(Pre-trained)

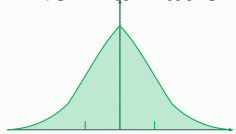


Semantic Embedding

Z_{LLM}



Z-Score
Normalization



3 Dual-Branch Autoencoder

Semantic Branch (LLM)

Normalized Embedding



LLM Branch
Encoder

Semantic Latent

H_Z



Attribute Branch

Attribute Branch



Attribute
Encoder

Attribute Latent

H_X



Gated Fusion

Fused Representation Z



4 Graph Anomaly Score

Shared Decoder

Reconstructed
Adjacency

$\hat{A}$



Reconstructed
Attributes

$\hat{X}$



Reconstruction Loss

$$s_i = (1 - \alpha) \|a_i - \hat{a}_i\|_2^2 + \alpha \|x_i - \hat{x}_i\|_2^2$$

Anomaly Score

s_i

Low



High